Jeyavinoth Jeyaratnam

jeyavinoth@gmail.com · (347) 307-7672 · jeyavinoth.com · linkedin.com/in/jeyavinoth

SUMMARY

Senior data engineer with 10+ years of experience working with messy external observational data: satellite telemetry, emissions inventories, weather signals, sensor networks, crowd-sourced reports. Most of the job is detective work, figuring out what unfamiliar data means and how to represent it before it can be useful downstream. Equally comfortable on the production ingestion side and on the exploration tooling research teams use day-to-day. Python, SQL, pandas, Bayesian methods, Dagster, Postgres.

EXPERIENCE

WattTime, Senior Data Engineer Apr 2022 – Present

Remote

- Built and deployed a Kubernetes-scheduled scraper fleet that pulls facility-generation data from 50+ sources in formats like PDF, Excel, CSV, XML, Parquet, HDF5, NetCDF, REST APIs, vendor-specific binary feeds, and unstructured text. Processed data lands in a normalized PostgreSQL table that modelers and analysts query directly, with a reliable monthly delivery cadence to Climate TRACE.
- Built the ingestion pipeline behind Climate TRACE's global emissions inventory, which processes roughly 1B records a year across 660M+ emitting assets in 200+ countries. End-to-end runtime fell from 10 days to 2; cloud compute cost cut in half.
- Worked out that hourly operation of power-plant cooling-tower fans could be inferred from RGB blur in commercial satellite imagery, and turned that into hourly CO_2 estimates for 1,000+ facilities. Combined it with satellite NO_2 and CO_2 observations to bring previously unmodeled capacity below 1,000 GW.
- Evaluated incoming external datasets from research partners at NASA, Stanford, and UW: decided what was worth ingesting, defined how it should sit in the warehouse, and got noisy raw observations into a state production research could actually run on.
- Refactored the satellite-image ingestion path to use async I/O and parallel processing: 10× throughput, 3 days to 6 hours. Migrated orchestration to Dagster for end-to-end lineage and run-level observability.
- Built the internal data-exploration tool the research team uses to validate model outputs against raw observations. A typical investigation that used to take an hour of hand-written SQL now takes under a minute.
- Pushed emissions-inference model accuracy up 20% by integrating plume-segmentation outputs and building the annotation tooling that connected external labelers to the internal label database.

The City College of New York, Senior Research Associate / PhD Candidate Sept 2018 – Apr 2022

New York, NY

- Reconciled two decades of inconsistent satellite humidity observations into a continuous climate data record by cross-validating three independent calibration methods against aircraft ground-truth profiles.
- Built a bi-level optimization to place a statewide network of roadside weather sensors, weighing storm severity and inter-storm variability against per-site installation cost. NYSDOT adopted the placement; estimated 18% annual saving on winter-maintenance labor and material.
- Built and maintained the group's SQL and MongoDB stores for multi-instrument satellite, radar, and model observations. Wrote the computer-vision front-detection algorithm used in two NASA diagnostic-tool components shipped via GitHub.
- Built a hurricane-intensity model that reached $R^2 = 0.9$ by combining satellite humidity profiles with NOAA hurricane-hunter reconnaissance data.

Jet Propulsion Laboratory, NASA, Data Science Intern May 2018 – Sep 2018

Pasadena, CA

- Built a Bayesian estimator that combined satellite infrared and microwave observations with a physics-based convection model to retrieve cloud vertical-velocity profiles. RMSE against ground-truth radar measurements fell 28%.
- Used random-forest feature importance to isolate the two observed quantities carrying most of the signal, directing the modeling team toward a smaller and more defensible feature set.

Lamont-Doherty Earth Observatory, Columbia University, Data Scientist Sep 2015 – May 2018

New York, NY

- Lead programmer on NASA's operational Snow Water Equivalent retrieval. Replaced static grain-size and density inputs with a neural-network grain-size estimator and dynamic density maps: SWE correlation against ground stations improved 25%, snow-temperature RMSE dropped 30%.
- Rewrote the snow-retrieval pipeline to read from a replacement satellite sensor after the primary instrument failed mid-mission, including the cross-instrument calibration needed to keep the multi-decade climate record continuous.
- Ran sensitivity analyses on regional climate models over Greenland and identified the persistent atmospheric ridge responsible for 2015's record poleward shift in melt, albedo, and surface temperature.

TECHNICAL SKILLS

Python (primary), SQL, pandas, NumPy, C/C++, MATLAB, shell. PostgreSQL, MongoDB, Redis Streams, SQLite. Airflow, Dagster, Docker, Kubernetes, Nginx. AWS, GCP. Statistical analysis, Bayesian methods, neural networks, computer vision, asynchronous and parallel data processing.

VENTURES & PROJECTS

CinchCourt LLC, Co-Founder (side venture; advisor since 2025) Jan 2025 – Dec 2025

Remote · operational role through 2025, advisory thereafter

- Built a trust-weighted Bayesian consensus that reconciles up to four conflicting reporter submissions per match into one confidence-weighted score, with per-reporter trust adapting from historical accuracy. The consensus confidence flows into the rating engine so noisier matches move ratings proportionally less.
- Cut matchmaking candidate evaluation from a brute-force $O(N^4)$ team enumeration to $\sim O(N \cdot K^3)$ by sorting players on rating and binary-searching a viable rating window, then scoring candidates on a weighted combination of team balance, rating spread, and wait time.
- Designed the analytics warehouse around a denormalized match-history fact table for low-latency leaderboard reads, with per-park materialized views over match-quality distributions feeding the dashboards. With that in place, matchmaking-policy changes could be evaluated against numbers rather than by feel.

Playable Weather, Hobby project (solo developer) Sep 2024 – Jan 2025

Multi-source weather and geospatial pipeline producing a per-court pickleball playability index

- Built a scraper that produces a complete metro-area pickleball-court inventory by recursively partitioning the search area into smaller bounding boxes to stay within per-query response caps, with polite rate-limiting and idempotent snapshots for safe retries.
- Cut hourly pipeline runtime from minutes to seconds: batched weather API calls into multi-location requests, de-duplicated forecast pulls when multiple courts mapped to the same grid cell, and cached the slow location-to-grid lookups so steady-state runs skipped them.
- Built a feature set with no public equivalent, derived from raw OpenStreetMap geometry: court orientation from each polygon's long axis, surface roughness from surrounding land-use tags, and an 8-direction shelter score from nearby tree and building footprints.
- Combined seven environmental sub-scores (wind, gust factor, thermal comfort, precipitation, surface wetness, air quality, light) into a single playability index. The composite uses a worst-factor-capped weighted average, so one bad factor still drops the score even if the other six look great.

EDUCATION

PhD candidate in Earth and Atmospheric Sciences, The Graduate Center, CUNY [4.0 GPA], Dissertation Pending (2027)

M.E. in Electrical Engineering, The City College of New York [3.9 GPA], 2014

B.E. in Electrical Engineering (Summa Cum Laude), The City College of New York [3.8 GPA], 2010

SELECTED PUBLICATIONS

First-author work in Geophysical Research Letters; co-author on papers in Nature Communications, Journal of Climate, PNAS, and Journal of Applied Meteorology and Climatology. Full list at jeyavinoth.com.